



Using Quantum Optics to Validate the Quantum Nature of Light

Russell Dillon

Introduction



After Einstein's paper on the photoelectric effect, people did not want to accept the quantum description of light. Semi-classical theory was invented to counter light being quantized.

Two experiments were created that showed light did not behave classically:

- Hanbury Brown-Twiss (HBT)
- Grangier-Rogers-Aspect (GRA)

Our goal is to recreate these experiments to test the quantum theory of light.

The Problem



We want to design an experiment that quantifies the behavior of light.

Figure A:
Classical light passing through a beamsplitter

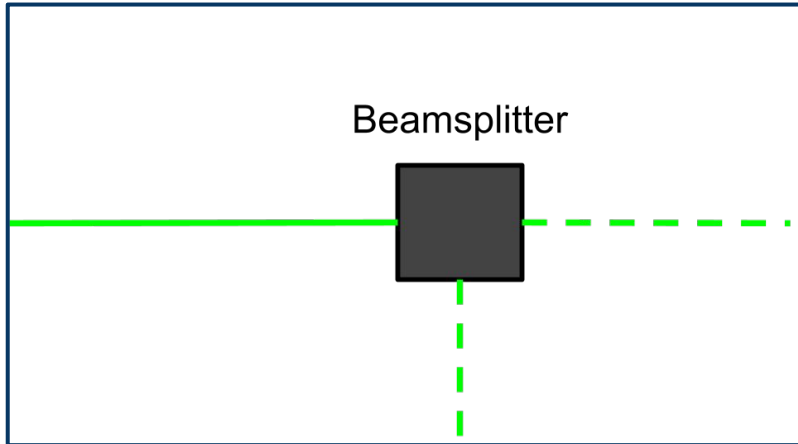
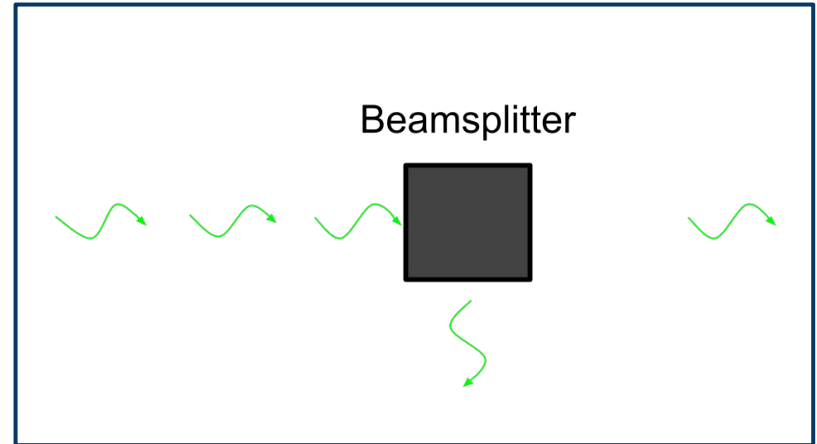


Figure B:
quantum light passing through a beamsplitter



The Problem Continued



To quantify this behavior, we will count the rate of photon detection on both sides of the beam splitter using the second order correlation functions:

$$g^2(0) = \frac{R_{AB}}{R_A R_B} \quad (1)$$

$$g^2(0) = \frac{R_{TAB} R_T}{R_{TA} R_{TB}} \quad (2)$$

Equations 1 and 2: second order correlation functions for HBT and GRA measurements, respectively. Values close to zero indicate the presence of single photons. Values far greater than zero indicate classical light.

Experimental Setups

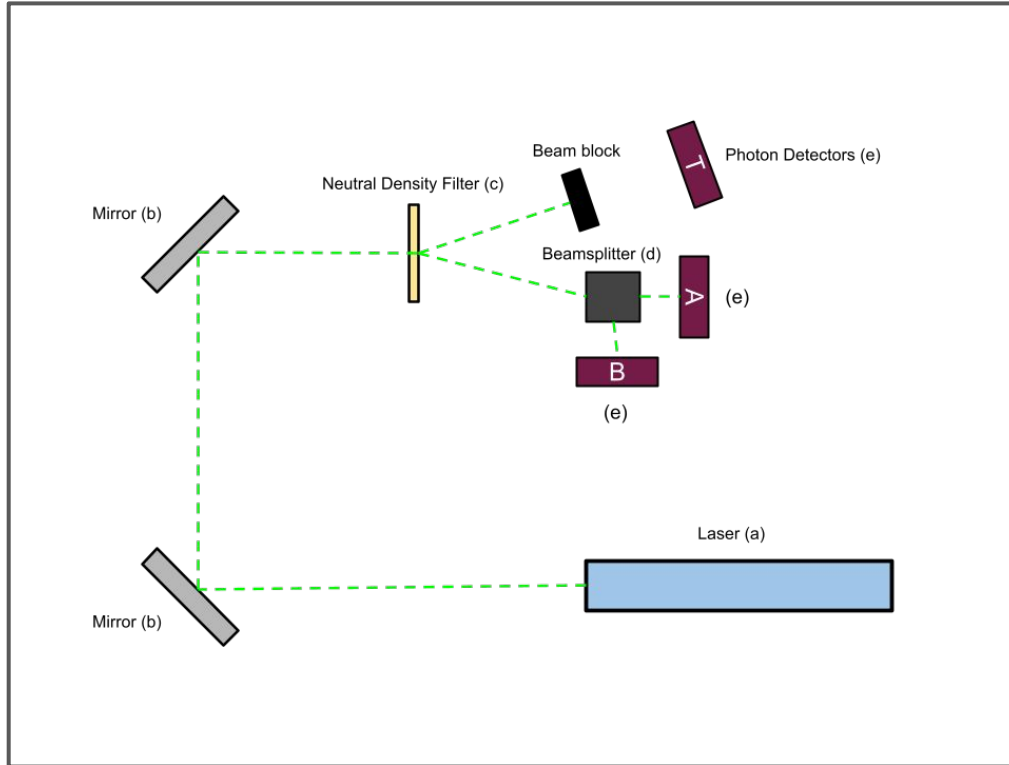


Diagram A: Experimental layout of first HBT measurement. Laser light is passed through a neutral density filter, towards two detectors A and B. The first HBT measurement yielded a $g^2(0)$ value of 1.02 ± 0.10 .

$$g^2(0) = \frac{R_{AB}}{R_A R_B} \quad (1)$$

Experimental Setups

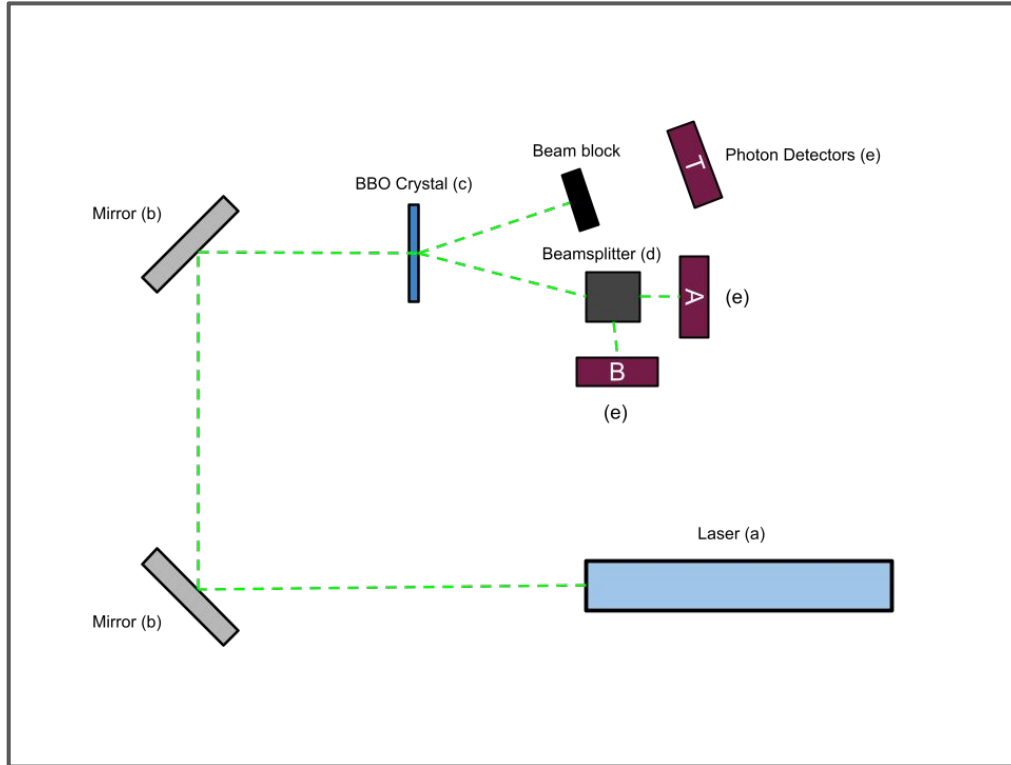


Diagram B: Experimental layout of second HBT measurement, modified so the neutral density filter is replaced by a beta-barium-borate (BBO) crystal. This experiment yielded a $g^2(0) = 1.22 \pm 0.12$.

$$g^2(0) = \frac{R_{AB}}{R_A R_B} \quad (1)$$

Experimental Setups

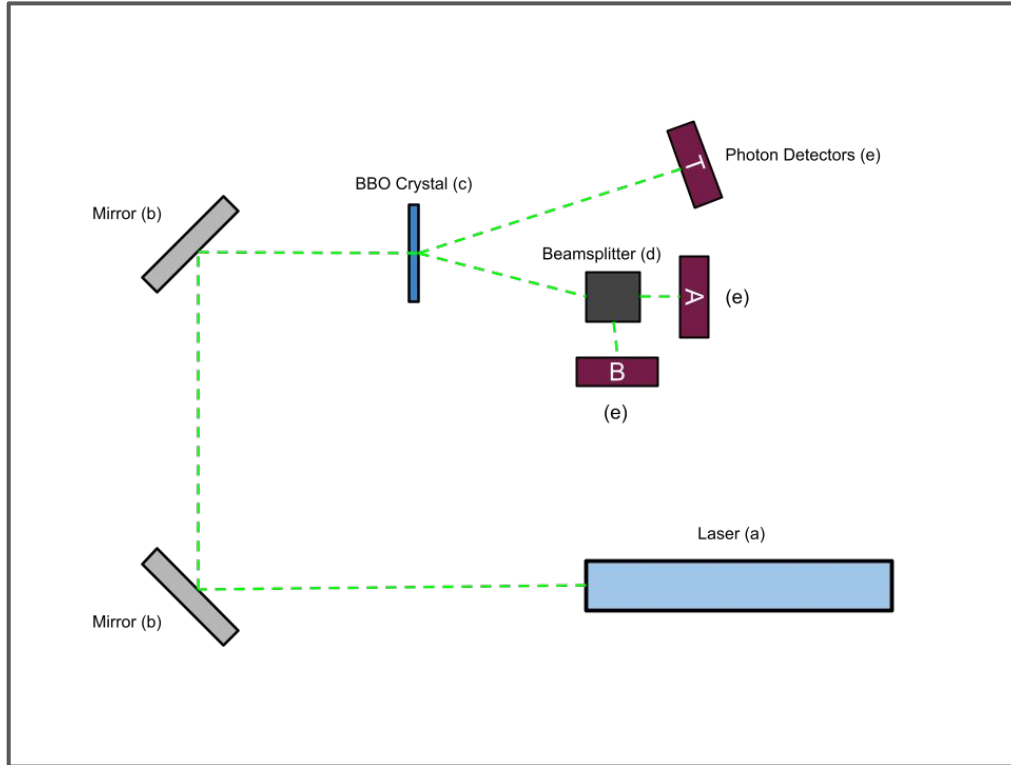


Diagram C: Experimental layout of the GRA measurement. We remove the beamblock and allow light to travel to all three detectors T, A, and B. We now use equation (2) and compute a $g^2(0) = (1.9 \pm 0.34) \cdot 10^{-3}$

$$g^2(0) = \frac{R_{TAB}R_T}{R_{TA}R_{TB}} \quad (2)$$

Uncertainty



We repeated each experiment ten times, making ten count rates for all detectors. The standard error of the mean (SEM):

$$SEM = \alpha_{R_X} = \frac{\sigma}{\sqrt{N}}$$

Across all ten trials was determined to be our count rate error.

We treated our second order correlation function as:

$$g^2(0) = \frac{R_{1eff}}{R_{2eff}}$$

And used the calculus based propagation:

$$\alpha_{g^2(0)} = g^2(0) \cdot \sqrt{\left[\frac{\alpha_{R_{1eff}}}{R_{1eff}}\right]^2 + \left[\frac{\alpha_{R_{2eff}}}{R_{2eff}}\right]^2} \quad (3)$$

Results

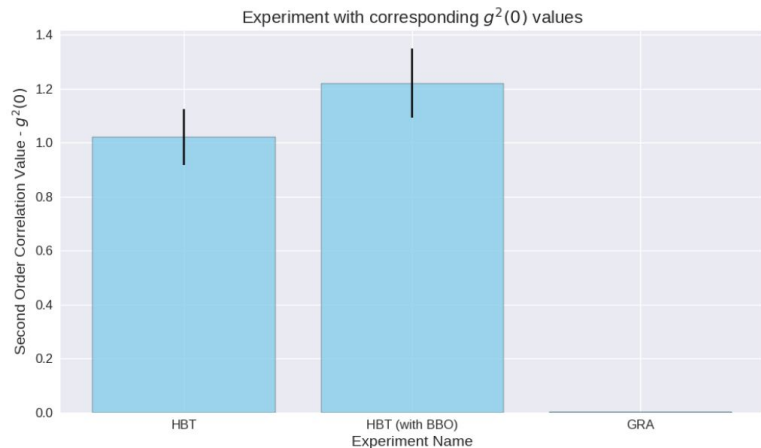


Figure 3: Histogram of experiment results across all three experiments: The HBT, HBT with BBO Crystal, and GRA experiments.

Experiment	$g^2(0)$	$\alpha_{g^2(0)}$
HBT	1.02	0.10
HBT (with BBO)	1.22	0.12
GRA	$1.9 \cdot 10^{-3}$	$3.4 \cdot 10^{-4}$

Figure 4: Table of second order correlation values, and corresponding uncertainties

Troubleshooting



Originally, there were four experiments. The fourth being a modification of the GRA experiment with a spectral filter replacing the BBO crystal. We noticed a zero simultaneous counts across three detectors after 10 trials. To fix we:

1. Aligned the Detectors
2. Re-aligned the laser
3. Aligned the spectral filter

This took a LONG time, and it is likely the issue is still laser misalignment.

Conclusion



Given the measured $g^2(0)$ values are consistent with theoretical expectations and historical measurements. Our recording of a correlation-value close to zero indicates the presence of single photons, supporting quantum theory as the framework to describe light's behavior.

Moving further, it would be wise to fix our scrapped experiment, and redo the GRA measurements with classical light. The realization of single photons has massive applications in modern physics. Solar panels, optical sensors, and quantum computing all benefit from quantum mechanics, and further research manipulating single photons could prove beneficial.



Thank you!
What are your questions?